

Chapter 1: Computer System

Overview of Hardware, Software, and Data Management

1.1 Introduction to Computer System

A computer is an electronic device programmed to accept data (input), process it, and generate a result (output). A computer system is the integration of the computer itself along with additional hardware and software required to perform tasks effectively.

The system is primarily composed of four main components: the Central Processing Unit (CPU), memory, input/output devices, and storage devices. These components work as a unified unit to deliver the desired output based on user instructions.

- A computer performs the Input-Process-Output (IPO) cycle.
- The computer system consists of both physical hardware and functional software.
- It can range in size from smartphones to high-end servers.

1.1.1 Central Processing Unit (CPU)

The CPU is the electronic circuitry that performs the actual data processing and is widely known as the 'brain' of the computer. It fetches instructions and data from the memory to perform arithmetic and logic operations. Physically, a CPU is a microprocessor placed on an Integrated Circuit (IC) made of semiconductor materials.

Inside the CPU, high-speed storage locations called registers are used to hold data, instructions, or intermediate results temporarily during execution. The size and number of these registers significantly impact the processor's speed.

- The CPU consists of the Arithmetic Logic Unit (ALU) and the Control Unit (CU).
- ALU handles mathematical calculations and logical comparisons.
- CU manages the flow of data and sequential execution of instructions.

1.1.2 1.1.3 Input and Output Devices

Input devices are used to send data and control signals to the computer. They convert human-readable data into a digital format (binary) that the computer can process. Common examples include keyboards, mice, and scanners.

Output devices receive digital data from the computer and convert it back into a human-understandable form, such as text on a monitor or a printed page. Advanced devices like 3D printers can even create physical replicas of digital designs.

- Input devices like Braille keyboards assist visually impaired users.
- Printers provide hard-copies, while monitors provide soft-copies.
- The 3D printer is used for manufacturing prototypes and medical organs.

1.2 Evolution of Computing

Computing has evolved from simple mechanical tools to complex electronic systems. Key milestones include the ABACUS (3000 years ago), Pascaline (1642), and Babbage's Analytical Engine (1834), which laid the foundation for modern computers.

The development of the Von Neumann architecture in 1945 introduced the concept of the 'stored-program computer,' where data and programs are kept in the same memory. Later, the invention of transistors and Integrated Circuits (ICs) allowed for the miniaturization and exponential growth of processing power.

- Von Neumann architecture consists of a CPU, memory, and I/O channels.
- Moore's Law predicted that the number of transistors on a chip doubles every 2 years.
- The shift from vacuum tubes to transistors revolutionized computer size and efficiency.

1.3.1 Units of Memory

Computers use the binary system (0 and 1) to store data. A single binary digit is called a bit. Groups of bits form larger units to represent more complex information. A group of 4 bits is a Nibble, and a group of 8 bits is a Byte.

Memory capacity is measured in multiples of 1024. For example, 1 *KB* is equal to 1024 *Bytes*. This power-of-two scaling continues up to Yottabytes (*YB*).

- Basic unit: 1 *Byte* = 8 *bits*.
- Standard scaling: 1 *GB* = 1024 *MB* and 1 *TB* = 1024 *GB*.
- Large scale units: 1 *PB* = 1024 *TB* and 1 *EB* = 1024 *PB*.

1.3.2 Types of Memory: Primary and Cache

Primary memory is essential for the computer to function. It is categorized into RAM and ROM. RAM (Random Access Memory) is volatile, meaning it loses data when power is off. It stores data temporarily for current tasks. ROM (Read Only Memory) is non-volatile and stores permanent startup instructions like the boot loader.

Cache memory is a very high-speed memory placed between the CPU and RAM. It stores frequently used data to reduce the average time the CPU takes to access main memory, thereby speeding up the system.

- RAM is volatile; ROM is non-volatile.
- The boot loader program is stored in ROM.
- Cache memory acts as a high-speed buffer for the processor.

1.3.2 (C) Secondary Memory

Secondary memory is used for permanent storage of data and instructions. It is non-volatile and has a much larger capacity than primary memory but is slower. Devices like Hard Disk Drives (HDD), Solid State Drives (SSD), and Pen Drives are common examples.

Unlike primary memory, the CPU cannot access secondary memory directly. Data must first be moved into the RAM before the CPU can process it. Modern SSDs are preferred over HDDs due to their higher data transfer speeds.

- Provides permanent storage for files and software.
- Examples include HDD, CD/DVD, and Memory Cards.
- SSD is faster and more reliable than traditional magnetic HDDs.

1.4 Data Transfer and System Bus

Data transfer between the CPU, memory, and I/O devices happens through physical wires called buses. These are categorized into three types: Data Bus, Address Bus, and Control Bus. Together, they are known as the System Bus.

The Data bus is bidirectional because data flows both into and out of the CPU. However, the Address bus and Control bus are unidirectional, as the CPU typically initiates the location and the type of operation (Read/Write).

- Data Bus: Carries actual data; Bidirectional.
- Address Bus: Specifies memory locations; Unidirectional.
- Control Bus: Carries signals like Read/Write; Unidirectional.

1.5 Microprocessors and Microcontrollers

A microprocessor is a CPU built on a single chip. Its performance is defined by Word Size (max bits processed at once), Clock Speed (pulses per second), and the number of Cores (computational units).

A microcontroller is a compact chip that contains a CPU, RAM, ROM, and I/O ports all in one. These are designed for specific tasks, such as controlling a washing machine or a microwave oven, where high-end general processing is not required.

- Modern processors have multiple cores (Dual, Quad, Octa-core).
- Microcontrollers are embedded systems for dedicated tasks.
- Clock speed is measured in Gigahertz (GHz).

1.6 Data Types and Management

Data consists of raw facts that are processed into meaningful information. It is categorized into three types: Structured (well-defined rows/columns like tables), Unstructured (no fixed format like videos or posts), and Semi-structured (uses tags but no rigid table, like HTML or CSV).

Data management involves capturing data from sources, storing it securely, and retrieving it for processing. As databases grow, minimizing retrieval time becomes a significant challenge for efficient computing.

- Structured data is easy to sort and search.
- Semi-structured data uses internal tags to separate elements.
- Data retrieval involves fetching data from storage for active use.

1.7 Categories of Software

Software is a set of instructions that makes hardware functional. It is classified into: System Software (provides basic functionality like OS), Programming Tools (translators and IDEs), and Application Software (performs specific user tasks).

Free and Open Source Software (FOSS) allows users to access and modify the source code (e.g., Ubuntu, Python). In contrast, Proprietary software is copyrighted and must be purchased (e.g., Windows, Tally).

- System software acts as an interface between user and hardware.
- Application software can be General-purpose or Customized.
- FOSS encourages collaborative development and improvement.

1.7.3 Programming Languages and Translators

Computers only understand machine language (0s and 1s). To make programming easier, High-Level Languages (HLL) like Python and Java were developed. These require translators to convert source code into machine-executable object code.

There are three types of translators: Assemblers (for assembly language), Compilers (translate whole program at once), and Interpreters (translate line-by-line).

- Compilers are not needed after the initial translation.
- Interpreters are required every time the source code is executed.
- IDEs like IDLE or Eclipse bundle editors and debuggers together.

1.8 Operating System (OS)

The Operating System is the most important system software. it acts as a resource manager that manages CPU time, memory space, and I/O devices. It also provides a User Interface (UI) through which a human interacts with the machine.

Modern OS types include Command-based (MS-DOS), Graphical (Windows/Ubuntu), Touch-based (Android/iOS), and newer Voice or Gesture-based interfaces.

- OS functions include Process, Memory, File, and Device management.
- GUI uses icons and menus; CLI requires text commands.
- The OS loads programs into RAM and allocates CPU cycles.

